Comptia N+ Assignment

[N+ Network Fundamental and building networks]

Introduction to Networking and the OSI Model

What is Networking?Overview of the OSI Model: Layers and their functions. Use a network sniffer (e.g., Wireshark) to observe traffic and identify layers.Task: Create a diagram of the OSI Model with examples for each layer.

Task

- **1. Install Wireshark on your laptop, start a capture on your main network interface, and visit any website (like instagram.com). Stop the capture and note down the protocol names you see in the packet list.**

  - **Aim**
  - To install and use Wireshark to capture network traffic, visit a website, and identify the different protocols observed in the captured packets.

  - **Introduction:**
  - A computer network is a collection of interconnected devices that communicate with each other and share data and resources.
  - When we visit a website, our computer communicates with different network devices and servers using various network protocols. These

protocols work at different layers of the OSI (Open Systems Interconnection) Model.

- Wireshark is a network protocol analyzer that allows us to capture and analyze packets travelling through a network interface.

- **Requirements:**
- The following are required for this practical:
  - A laptop or computer
  - Internet connection
  - Wireshark software
  - A web browser
  - A website such as instagram.com

- **Procedure:**
- **Step 1: Install Wireshark:**
  - Download and install Wireshark on the laptop.
  - Open Wireshark after installation.
  - Wireshark displays the available network interfaces.

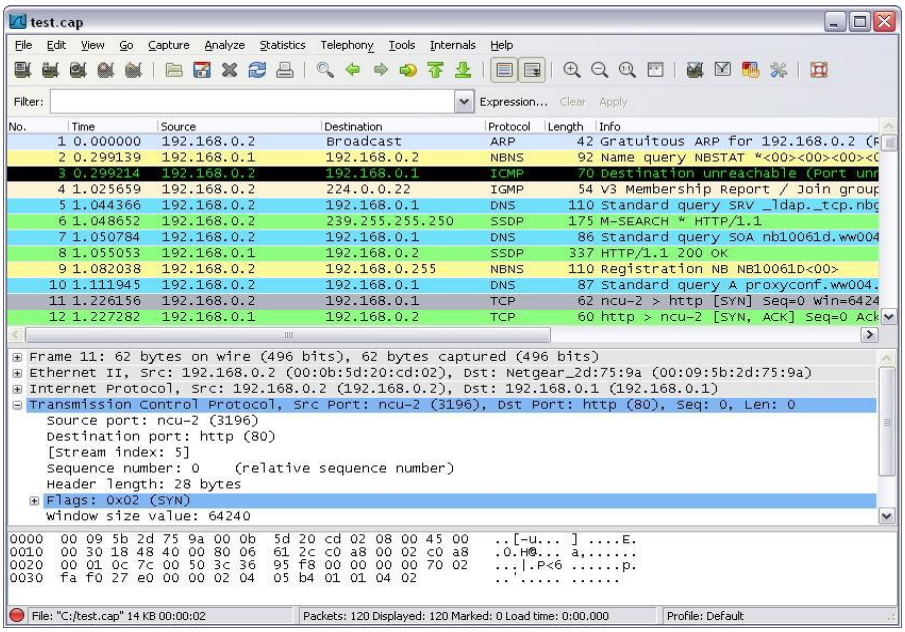
- **Step 2: Select the Network Interface:**
  - Identify the network interface currently being used.
  - If connected through Wi-Fi, select the Wi-Fi interface.
  - If connected through a LAN cable, select the Ethernet interface.
  - Start capturing packets by double-clicking the active interface.

- **Step 3: Start Packet Capture:**

- After starting the capture, Wireshark begins displaying network packets in real time.
- The packet list generally contains information such as:
  - Packet Number
  - Time
  - Source
  - Destination
  - Protocol
  - Length
  - Information

- **Step 4: Visit a Website**
  - Open a web browser.
  - Visit a website such as **instagram.com**.
  - Allow the website to load completely.
  - During this process, Wireshark captures the network packets generated by the communication.

- **Step 5: Stop the Capture**
  - Return to Wireshark.
  - Click the **Stop** button.
  - The captured packets can now be examined.

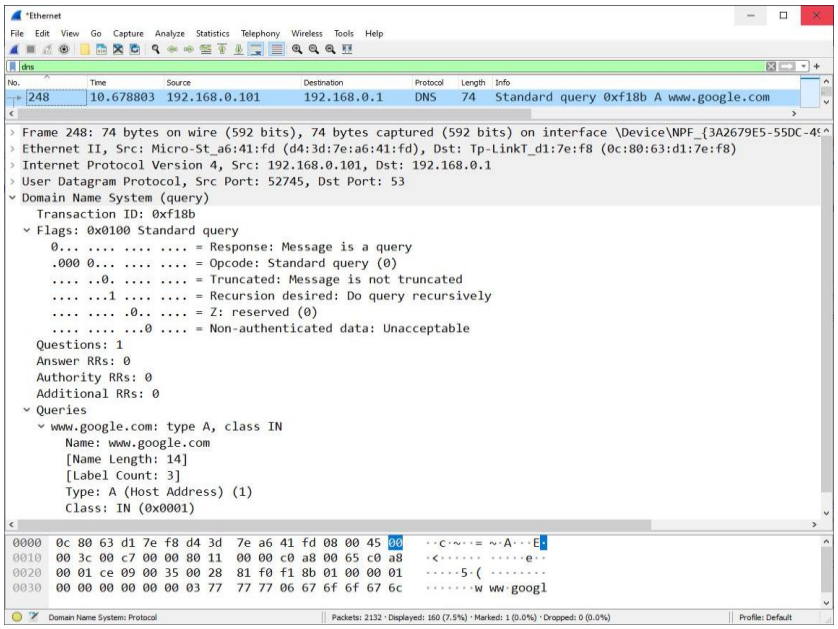


➢ Observation

- After visiting the website, different protocols can be observed in the Wireshark packet list.
- Common protocols that may be visible**:**

Sr No.	Protocol	Purpose
1	DNS	Converts a domain name into IP address.
2	TCP	Provides reliable communication between devices.

3	TLS	Provides encryption and security for communication.
4	HTTP	Used for transferring web resources when HTTP is used.
5	QUIC	Provides fast and secure transport, commonly used with HTTP/3.
6	UDP	Provides connectionless transport communication.
7	ARP	Helps determine the MAC address associated with an IPv4 address on a local network.
8	ICMP	Used for network control and diagnostic messages.



➢ Relation with the OSI Model

- The observed network protocols can be related to the OSI Model

Layer	Name	Function	Example Protocols / Technologies
7	Application	Provides network services to end-user applications.	HTTP, HTTPS, FTP, DNS, SMTP, IMAP
6	Presentation	Handles data format, encryption and compression.	TLS/SSL, JPEG, MPEG, ASCII
5	Session	Manages sessions and connection control.	NetBIOS, RPC, PPTP
4	Transport	Provides end-to-end communication and reliability.	TCP, UDP, SCTP
3	Network	Handles logical addressing and routing.	IP, ICMP, IGMP, IPsec
2	Data Link	Handles framing, MAC addressing and error detection.	Ethernet, Wi-Fi, ARP, PPP
1	Physical	Transmits raw bits over the physical medium.	Ethernet cable, Wi-Fi signals, Fiber, Bluetooth

- **How Wireshark Helps in Understanding the OSI Model**
  - Wireshark provides detailed information about captured packets.
  - By selecting a packet, we can expand its protocol details and observe information such as:
  - Ethernet frame
  - Source and destination MAC addresses
  - IP addresses
  - TCP or UDP information
  - DNS information
  - TLS information
  - Application-level information

- **Result**
  - The Wireshark practical was successfully performed. Network traffic was captured while visiting a website, and different protocols such as **DNS, TCP, TLS, UDP, QUIC, ARP and IP-related traffic** could be observed in the packet list.
  - This practical demonstrates how different networking protocols work together to provide communication between a user's computer and Internet servers.

❖ **2. Pick any 3 packets from your Wireshark capture and identify which OSI layer each packet mainly represents (for example, HTTP for Application layer, TCP for Transport layer, etc.). Use the Protocol column and Info details in Wireshark to help you map protocols to OSI layers.**

➢ **Aim**

- To select any three packets from a Wireshark network capture and identify the OSI layer mainly represented by each packet using the Protocol column and Info details.

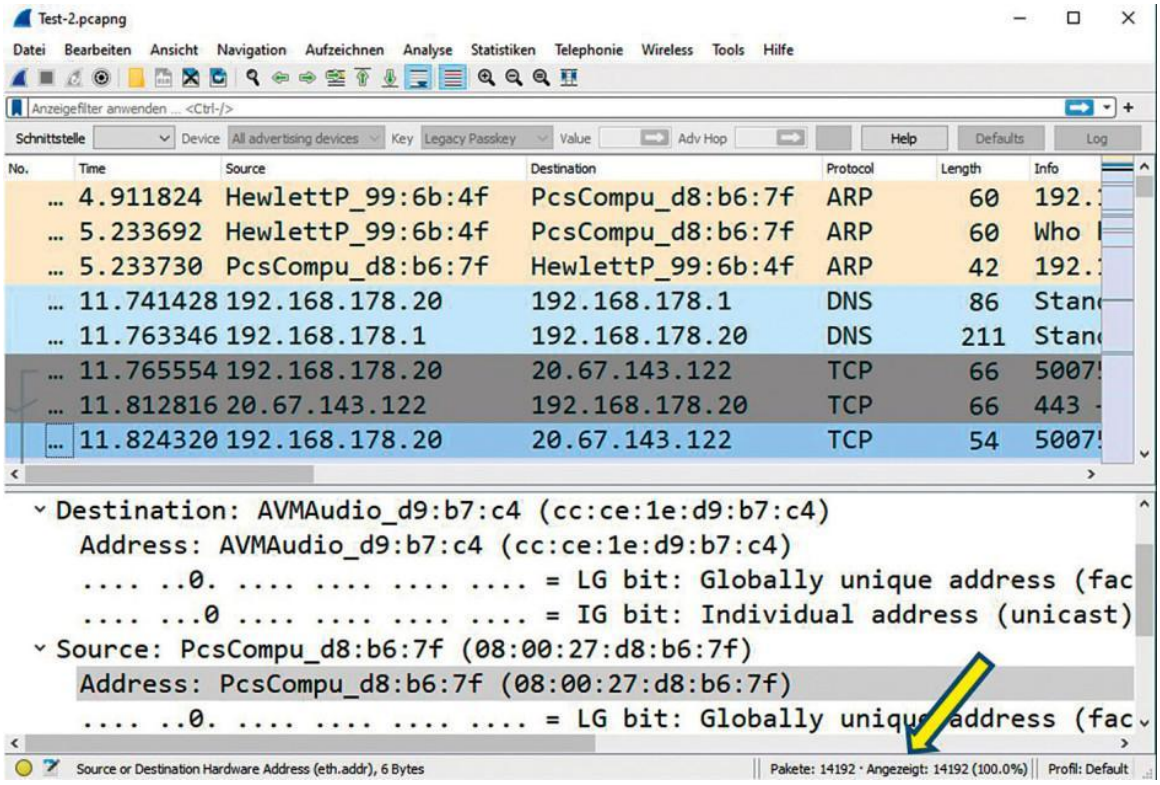
➢ **Introduction**

- Wireshark is a network protocol analyzer used to capture and examine network packets. Each packet contains information that helps us understand how data travels through a computer network.
- The OSI (Open Systems Interconnection) Model consists of seven layers. Different networking protocols operate at different layers of this model.
- By examining the **Protocol** and **Info** columns in Wireshark, we can identify the protocols used in network communication and map them to their corresponding OSI layers.

➢ **Procedure**

1. Open Wireshark and start capturing packets on the active network interface.
2. Open a website, such as instagram.com, in a web browser.
3. Stop the packet capture after the website loads.
4. Observe the Protocol and Info columns in the packet list.
5. Select three packets that contain identifiable protocols.

6. Examine their details and identify the corresponding OSI layers.
7. Record the observations in a table.



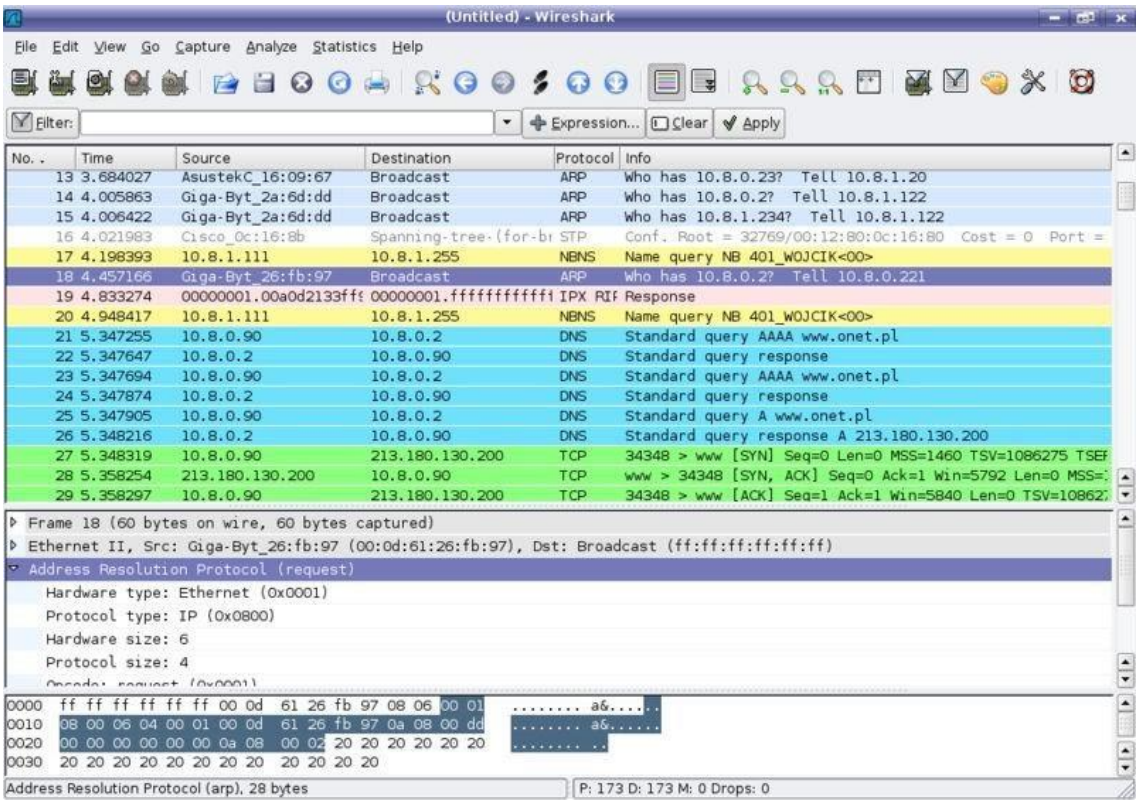
➢ Observation Table

- The following table shows an example of how three packets can be identified and mapped to the OSI Model.

Packet	**Protocol**	**Information observed in Wireshark**	**OSI Layer**
Packet 1	DNS	Standard query for a domain name,	Layer 7 – Application

		such as Instagram.com	
Packet 2	TCP	Connection establishment, such as SYN or ACK flags	Layer 4 – Transport
Packet 3	ARP	Packet Request to find the MAC address associated with an IPv4 address	Layer 2 – Data Link

- **Note:** These are example observations. The actual packet numbers and protocol names may differ in your capture. If ARP is not present, select another packet that is available, such as IPv4 or TLS, and map it to the appropriate layer.



- **Explanation of the Selected Packets**
- **Packet 1: DNS — Application Layer (Layer 7)**
  - DNS stands for Domain Name System. It translates human-readable domain names, such as instagram.com, into IP addresses used for network communication.
  - In Wireshark, the Protocol column may display DNS, while the Info column may show a standard query or response for a domain name.
  - Therefore, DNS mainly represents the **Application Layer (Layer 7)** of the OSI Model because it provides a network service to applications.
- **Packet 2: TCP — Transport Layer (Layer 4)**

- TCP stands for Transmission Control Protocol. It provides reliable, connection-oriented communication between devices.
- In Wireshark, the Protocol column may display TCP, while the Info column may contain details such as SYN, ACK, or source and destination port numbers.
- These details help identify TCP connection establishment and communication.
- Therefore, TCP represents the **Transport Layer (Layer 4)** because it manages end-to-end data delivery, sequencing, and reliability.

➢ **Packet 3: ARP — Data Link Layer (Layer 2)**

- ARP stands for Address Resolution Protocol. It is used in IPv4 local networks to discover the MAC address associated with a known IPv4 address.
- In Wireshark, the Protocol column may display ARP, while the Info column may show an ARP request or reply.
- ARP helps devices communicate over a local network by resolving IP addresses to MAC addresses.
- Therefore, ARP is commonly associated with the **Data Link Layer (Layer 2)** in the OSI Model.

➢ **Importance of Protocol and Info Columns**

- The Protocol column identifies the protocol associated with a packet. The Info column provides additional details about the packet, such as DNS queries, TCP flags, or ARP requests.
- By examining these two columns and expanding the packet details, we can identify the protocols and understand their roles in network communication.

- **Result**
  - Three network packets were examined using Wireshark, and their protocols were mapped to the corresponding OSI layers. DNS was associated with the Application Layer, TCP with the Transport Layer, and ARP with the Data Link Layer.

❖ **3. Create a colorful diagram of the OSI Model using any drawing tool (MS Paint, Canva, or even paper and upload a photo). For each layer, add a real-world example from apps you use daily (e.g., WhatsApp uses the Application layer, Instagram images use the Presentation layer, etc.)**

➢ **Aim**

- To create a colorful diagram of the OSI Model and explain each of its seven layers with real-world examples from applications and devices used in daily life.

➢ **Introduction**

- The OSI (Open Systems Interconnection) Model is a reference model that explains how data travels from one device to another over a computer network.
- It consists of seven layers, and each layer performs a specific function in network communication. These layers work together to help applications and devices exchange data efficiently.

➢ **Seven Layers of the OSI Model with Real-World Examples**

➢ **Layer 7: Application Layer**

- The Application Layer provides network services to applications used by end users. It allows applications to access network resources and communicate over the Internet.
- **Real-world example:** WhatsApp uses network protocols to send messages, make calls, and share media.
- **Protocols:** HTTP, HTTPS, DNS, SMTP.

- **Layer 6: Presentation Layer**
  - The Presentation Layer deals with data formatting, translation, encryption, decryption, and compression. It helps ensure that data is represented in a suitable format.
  - **Real-world example:** When we share images or videos on Instagram, the media is encoded, formatted, and may be compressed before transmission. Secure communication may also involve encryption.
  - **Examples:** JPEG, PNG, ASCII, encryption and compression formats.

- **Layer 5: Session Layer**
  - The Session Layer establishes, manages, and terminates communication sessions between applications.
  - **Real-world example:** During a Google Meet video call, the communication session is maintained while participants exchange audio and video.
  - **Examples:** Session management, RPC, NetBIOS.

- **Layer 4: Transport Layer**
  - The Transport Layer provides end-to-end communication between devices. It handles data segmentation, port numbers, and, depending on the protocol, reliability and flow control.
  - **Real-world example:** When watching a YouTube video, transport protocols carry data between the device and the server.
  - **Protocols:** TCP and UDP.

- **Layer 3: Network Layer**

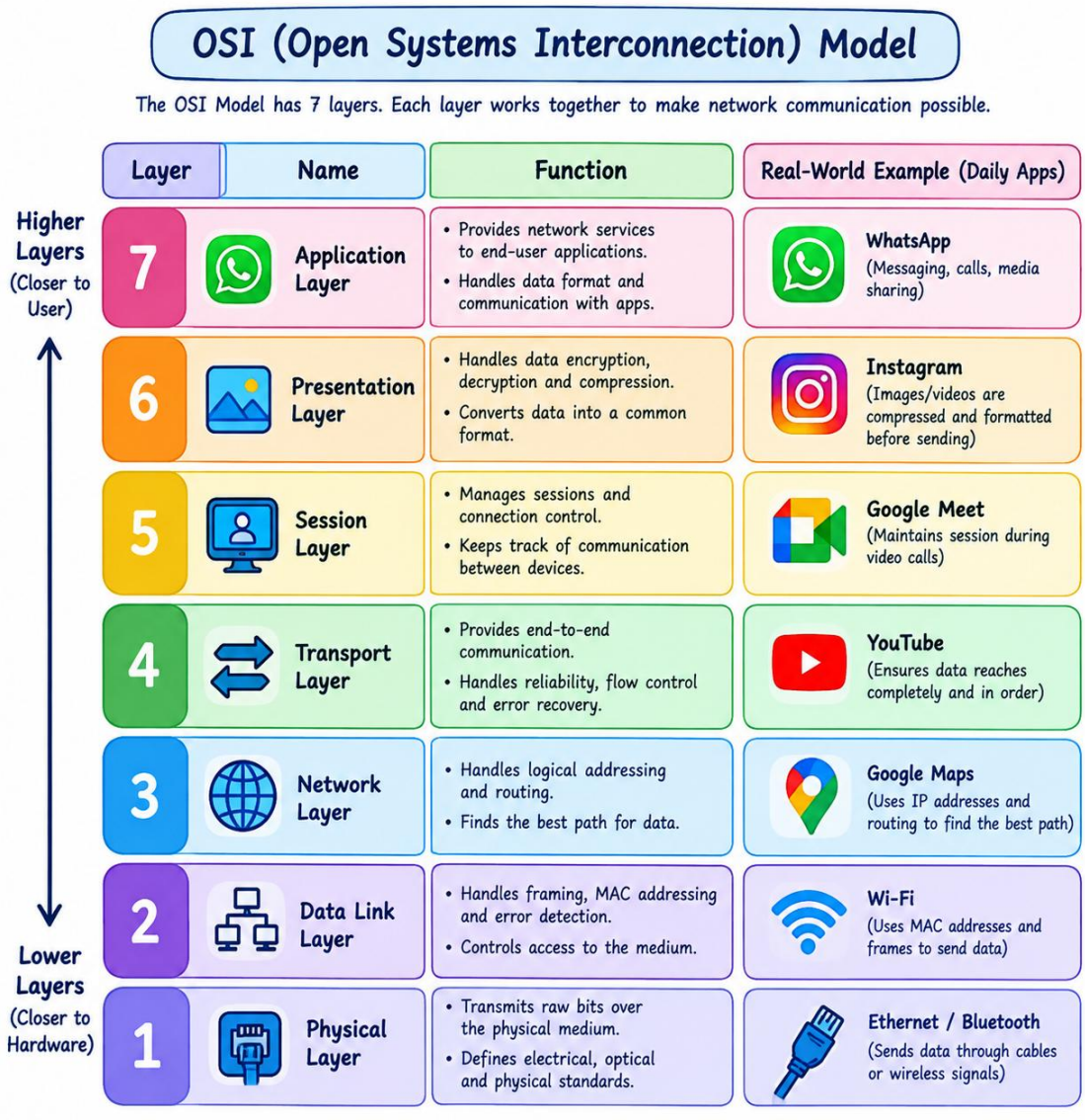
- The Network Layer handles logical addressing and routing. It determines how packets can travel from a source network to a destination network.
- **Real-world example:** Google Maps uses network communication to retrieve map information and directions from online services.
- **Protocols:** IP, ICMP.

➢ **Layer 2: Data Link Layer**

- The Data Link Layer transfers data in frames over a local network. It handles MAC addressing and access to the network medium.
- **Real-world example:** A laptop connected to a Wi-Fi router uses data-link functions to communicate over the local network.
- **Examples:** Ethernet, Wi-Fi (IEEE 802.11), PPP.

➢ **Layer 1: Physical Layer**

- The Physical Layer transmits raw bits through a physical medium or wireless signals. It defines the electrical, optical, and physical characteristics of the connection.
- **Real-world example:** An Ethernet cable carries network signals between a computer and a router. Wi-Fi uses radio signals to transmit data wirelessly.



(Colorful OSI Model with Seven Layers and Real-World Examples)

❖ **4. Explain how sending a message on WhatsApp travels through at least 4 layers of the OSI Model, describing what happens at each layer in your own words. Think about how your message is typed, packaged, sent over the internet, and received.**

➢ **Aim**

➢ To explain how a message sent through WhatsApp travels through the different layers of the OSI Model, from the sender's device to the receiver's device.

➢ **Introduction**

- WhatsApp is a messaging application that allows users to send text messages, images, videos, voice messages, and documents over the Internet.
- When we send a message on WhatsApp, the data passes through different networking processes before reaching the receiver. The OSI (Open Systems Interconnection) Model helps us understand these processes by dividing network communication into seven layers.
- Each layer performs a specific function to ensure that the message can be transmitted from one device to another.

➢ **Working of a WhatsApp Message Through the OSI Model**

- Let us understand the process with an example. Suppose a sender types **"Hello, how are you?"** on WhatsApp and sends it to a friend.

➢ **Layer 7: Application Layer**

- The Application Layer provides network services to applications. When the sender types a message and presses the Send button, WhatsApp prepares the message for transmission.
- **Example:** The sender types "Hello, how are you?" in the WhatsApp chat and presses Send.

➢ **Layer 6: Presentation Layer**

- The Presentation Layer deals with data representation, encryption, and compression. WhatsApp uses end-to-end encryption to protect messages so that only the intended participants' devices can read their contents.
- **Example:** The message is protected using WhatsApp's encryption system before it is transmitted. The actual implementation does not correspond exactly to one OSI layer.

➢ **Layer 5: Session Layer**

- The Session Layer is responsible for establishing, managing, and ending communication sessions between applications.
- **Example:** WhatsApp maintains the communication needed to exchange messages between the sender and receiver. In real networks, these session-related functions are handled by several protocols and software components rather than one separate Session Layer.

➢ **Layer 4: Transport Layer**

- The Transport Layer provides communication between devices or applications. It uses transport protocols and port numbers to carry data. Depending on the protocol, it may also provide reliability and error recovery.

- **Example:** Transport protocols carry the encrypted WhatsApp data towards the appropriate network destination.

➢ **Layer 3: Network Layer**

- The Network Layer handles logical addressing and routing. It uses IP addresses to move packets across different networks towards their destinations.
- **Example:** The sender's phone sends IP packets through the Internet towards the relevant WhatsApp service.

➢ **Layer 2: Data Link Layer**

- The Data Link Layer transfers data in frames over the local network. It uses link-layer addressing, such as MAC addresses on Ethernet or Wi-Fi, to deliver frames across the local connection.
- **Example:** When the phone is connected to Wi-Fi, the data is transmitted in Wi-Fi frames to the wireless router.

➢ **Layer 1: Physical Layer**

- The Physical Layer transmits raw bits using physical or wireless signals.
- **Example:** The message data travels as radio signals between the phone and the Wi-Fi router, or through the cellular radio connection when mobile data is used.

➢ **How the Message Reaches the Receiver**

- The sender types a message and presses Send in WhatsApp.
- WhatsApp prepares the message and protects its contents using end-to-end encryption.

- The data is carried through transport and network protocols.
- The data travels through the sender's local network and across the Internet to WhatsApp's service infrastructure.
- The service forwards the protected message towards the receiver when possible.
- The receiver's device obtains the data and processes it through the relevant networking layers.
- WhatsApp decrypts the message on the receiver's device and displays the original text.

➢ **Diagram of WhatsApp Message Transmission**
➢ **Sender's Phone → Internet → Receiver's Phone**
➢ SENDER'S PHONE
➢ |
➢ v
➢ ---
➢ | 7. Application Layer | WhatsApp message
➢ ---
➢ | 6. Presentation Layer | Encryption / formatting
➢ ---
➢ | 5. Session Layer | Communication management
➢ ---
➢ | 4. Transport Layer | Transport protocols
➢ ---
➢ | 3. Network Layer | IP addressing and routing
➢ ---
➢ | 2. Data Link Layer | Wi-Fi / cellular link frames

- ------------------------
- | 1. Physical Layer | Radio or cable signals
- ------------------------
- |
- v
- INTERNET / NETWORK
- |
- v
- RECEIVER'S PHONE
- Data is received,
- processed and displayed
- **Note:** This diagram is a simplified explanation of the OSI Model. Real-world networking protocols do not always map perfectly to individual OSI layers.

- **Conclusion**
  - Sending a WhatsApp message involves several networking functions. The Application Layer handles the messaging service, encryption protects the message contents, transport protocols carry the data, the Network Layer routes packets, and the Data Link and Physical Layers transmit data over local network connections.
  - At the receiver's end, the data is processed and the original message is displayed. Thus, the OSI Model helps us understand how a simple WhatsApp message travels from one device to another through the Internet.